

Course: BCSNT 6063 – Object Oriented Programming (2nd Semester)

Assignment Type: Practical / Tutorial

Submission: Screenshots + Program Files

1)StudentSystem.java

```
class Student {    private
String name;    private
int rollNumber;
    private String faculty;

    // Constructor
    public Student(String name, int rollNumber, String faculty)
{        this.name = name;        this.rollNumber = rollNumber;
        this.faculty = faculty;
    }

    // Getters    public
String getName() {
    return name;
}

    public int getRollNumber() {
        return rollNumber;
    }

    public String getFaculty() {
        return faculty;
    }

    // Setters
    public void setName(String name) {
this.name = name;
    }

    public void setRollNumber(int rollNumber) {
        this.rollNumber = rollNumber;
    }

    public void setFaculty(String faculty) {
        this.faculty = faculty;
    }

    // Display student profile
    public void displayProfile() {
```

```

System.out.println("----- Student
Admission Details -----");
    System.out.println("Name: " + name);
    System.out.println("Roll Number: " + rollNumber);
    System.out.println("Faculty: " + faculty);
}
}

public class StudentSystem {
    public static void main(String[] args) {

        Student s1 = new Student("Sumit khatri", 31, "Computer Science");

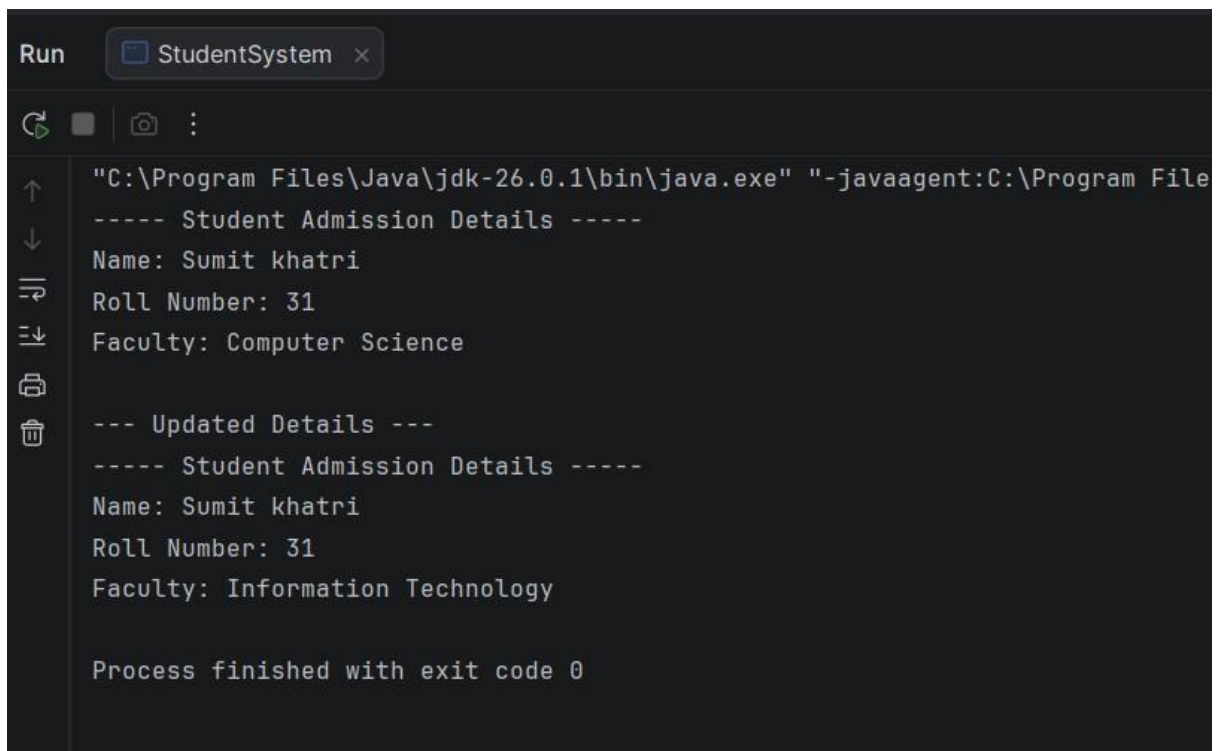
        s1.displayProfile();

        // Example update using setter
        s1.setFaculty("Information Technology");

        System.out.println("\n--- Updated Details ---");
        s1.displayProfile();
    }
}

```

Output



```

Run StudentSystem x
C:\Program Files\Java\jdk-26.0.1\bin\java.exe "-javaagent:C:\Program File
----- Student Admission Details -----
Name: Sumit khatri
Roll Number: 31
Faculty: Computer Science

--- Updated Details ---
----- Student Admission Details -----
Name: Sumit khatri
Roll Number: 31
Faculty: Information Technology

Process finished with exit code 0

```

2) EmployeeSystem.java

```
class Employee {
    String name;
    int id;

    public Employee(String name, int id) {
        this.name = name;    this.id = id;
    }

    public void displayDetails() {
        System.out.println("Employee Name: " + name);
        System.out.println("Employee ID: " + id);
    }
}

class Teacher extends Employee {
    String subject;

    public Teacher(String name, int id, String subject) {
        super(name, id);
        this.subject = subject;
    }

    @Override
    public void displayDetails() {
        System.out.println("----- Teaching Staff -----");
        super.displayDetails();
        System.out.println("Subject: " + subject);
    }
}

class AdminStaff extends Employee {
    String department;

    public AdminStaff(String name, int id, String department) {
        super(name, id);
        this.department = department;
    }

    @Override
    public void displayDetails() {
        System.out.println("----- Administrative Staff -----");
        super.displayDetails();
        System.out.println("Department: " + department);
    }
}
```

```
}
```

```
public class EmployeeSystem {  
    public static void main(String[] args) {  
  
        Teacher t1 = new Teacher("Ramesh Rai", 201, "Mathematics");  
        AdminStaff a1 = new AdminStaff("Gautam Sha", 301, "Finance");  
  
        t1.displayDetails();  
        System.out.println();  
  
        a1.displayDetails();  
    }  
}
```

Output

```
Run EmployeeSystem x  
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\  
----- Teaching Staff -----  
Employee Name: Ramesh Rai  
Employee ID: 201  
Subject: Mathematics  
----- Administrative Staff -----  
Employee Name: Gautam Sha  
Employee ID: 301  
Department: Finance  
  
Process finished with exit code 0
```

3) StudentCourseSystemQ4.java

```
import java.util.ArrayList;
```

```
class Course {  
    String courseName;  
  
    Course(String courseName) {  
        this.courseName = courseName;  
    }  
}
```

```

    }

    void displayCourse() {
        System.out.println(courseName);
    }
}

class StudentQ4 {
    String name;    int
    rollNumber;
    ArrayList<Course> courseList = new ArrayList<>();

    StudentQ4(String name, int rollNumber) {
        this.name = name;
        this.rollNumber = rollNumber;
    }

    void registerCourse(Course c) {
        courseList.add(c);
    }

    void displayRegisteredCourses() {
        System.out.println("Student Name: " + name);
        System.out.println("Roll Number: " + rollNumber);

        System.out.println("Registered Courses:");

        if (courseList.isEmpty()) {
            System.out.println("No courses registered.");
        }
        return;

        for (Course c : courseList) {
            c.displayCourse();
        }
    }
}

public class StudentCourseSystemQ4 {
    public static void main(String[] args) {

        StudentQ4 s1 = new StudentQ4("Ram Sharma", 101);

        // Register multiple courses
        s1.registerCourse(new Course("Mathematics"));
        s1.registerCourse(new Course("Physics"));
        s1.registerCourse(new Course("Computer Science"));
    }
}

```

```
// Display output
s1.displayRegisteredCourses();
}}
```

Output

The screenshot shows the 'Run' window of an IDE. The title bar reads 'StudentCourseSystemQ4'. The console output displays the command used to run the program, followed by the application's output. The output lists student details and registered courses.

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\Java\jdk-26.0.1\lib\jawsdk.jar" -Djconsole.port=7090 StudentCourseSystemQ4  
Student Name: Ram Sharma  
Roll Number: 101  
Registered Courses:  
Mathematics  
Physics  
Computer Science  
  
Process finished with exit code 0
```

4)AttendaceSystemQ7.java

```
class Attendance {
    String name;    int
    totalClasses;
        int attendedClasses;

    Attendance(String name, int totalClasses, int attendedClasses)
    {
        this.name = name;        this.totalClasses = totalClasses;
        this.attendedClasses = attendedClasses;
    }
}
```

```

    double calculateAttendance() {
return 0.0;
    }

    void displayInfo() {
        System.out.println("Name: " + name);
    }
}

class EngineeringAttendance extends Attendance {

    EngineeringAttendance(String name, int totalClasses, int attendedClasses) {
super(name, totalClasses, attendedClasses);
    }

    @Override
    double calculateAttendance() {
        return (attendedClasses * 100.0) / totalClasses;
    }

    @Override    void
displayInfo() {
        System.out.println("----- Engineering Faculty -----");
        super.displayInfo();
        System.out.println("Attendance: " + calculateAttendance() + "%");
    }
}

class MedicalAttendance extends Attendance {

    MedicalAttendance(String name, int totalClasses, int attendedClasses) {
super(name, totalClasses, attendedClasses);
    }

    @Override
    double calculateAttendance() {
        return (attendedClasses * 100.0) / totalClasses;
    }

    @Override    void
displayInfo() {
        System.out.println("----- Medical Faculty -----");
        super.displayInfo();
        System.out.println("Attendance: " + calculateAttendance() + "%");
    }
}

```

```

public class AttendanceSystemQ7 {
    public static void main(String[] args) {

        EngineeringAttendance e1 =
            new EngineeringAttendance("Ram Sharma", 100, 85);

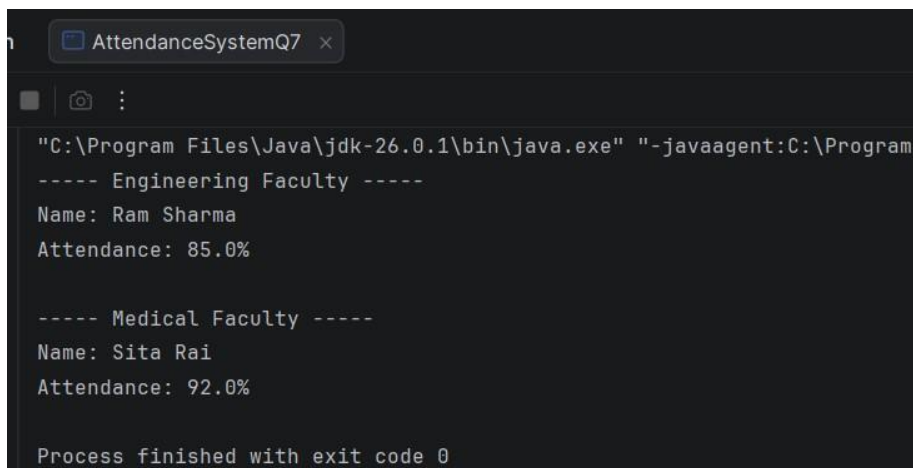
        MedicalAttendance m1 =
            new MedicalAttendance("Sita Rai", 100, 92);

        e1.displayInfo();
        System.out.println();

        m1.displayInfo();
    }
}

```

Output



```

AttendanceSystemQ7 x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program
----- Engineering Faculty -----
Name: Ram Sharma
Attendance: 85.0%

----- Medical Faculty -----
Name: Sita Rai
Attendance: 92.0%

Process finished with exit code 0

```

5) NotificationSystemQ8.java

```

interface Notification {
    void send(String message);
}

class EmailNotification implements Notification {

    @Override
    public void send(String message) {
        System.out.println("Sending EMAIL notification:");
        System.out.println("Message: " + message);
    }
}

```



```

class SMSNotification implements Notification {

    @Override
    public void send(String message) {
        System.out.println("Sending SMS notification:");
        System.out.println("Message: " + message);
    }
}

public class NotificationSystemQ8 {
    public static void main(String[] args) {

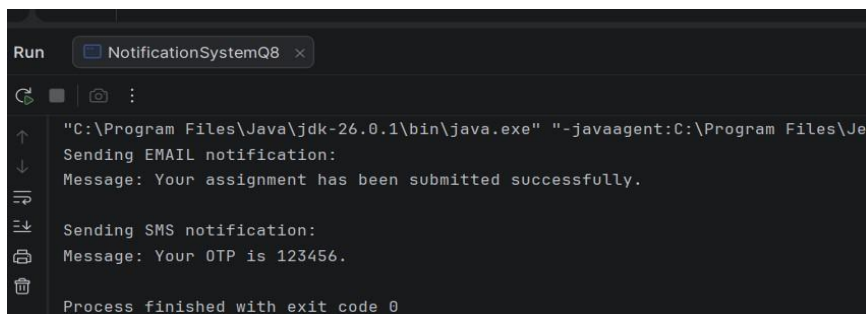
        Notification email = new EmailNotification();
        Notification sms = new SMSNotification();

        email.send("Your assignment has been submitted successfully.");
        System.out.println();

        sms.send("Your OTP is 123456.");
    }
}

```

Output



```

Run NotificationSystemQ8 x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\Jet
Sending EMAIL notification:
Message: Your assignment has been submitted successfully.

Sending SMS notification:
Message: Your OTP is 123456.

Process finished with exit code 0

```

6) StudentIdSystemQ9.java

```

class StudentQ9 {
    String name;
    int id;

    static int counter = 1000;

    StudentQ9(String name) {
        this.name = name;

        counter++;
        this.id = counter;
    }

    void display() {

```

```

        System.out.println("Name: " + name);
        System.out.println("Student ID: " + id);
        System.out.println();
    }
}

public class StudentIDSystemQ9 {
    public static void main(String[] args) {

        StudentQ9 s1 = new StudentQ9("Ram");
        StudentQ9 s2 = new StudentQ9("Sita");
        StudentQ9 s3 = new StudentQ9("Aayush");

        s1.display();
        s2.display();
        s3.display();
    }
}

```

Output



```

Run StudentIDSystemQ9 x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBr
Name: Ram
Student ID: 1001

Name: Sita
Student ID: 1002

Name: Aayush
Student ID: 1003

Process finished with exit code 0

```

7) ScholarshipSystemQ10.java

```

abstract class Scholarship {

    String name;
    double marks;

    Scholarship(String name, double marks) {
        this.name = name;
        this.marks = marks;
    }

    abstract void checkEligibility();

    void displayInfo() {
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }
}

```

```
}  
}
```

```
class MeritScholarship extends Scholarship {
```

```
    MeritScholarship(String name, double marks) {  
        super(name, marks);  
    }
```

```
    @Override  
    void checkEligibility() {  
        System.out.println("----- Merit Scholarship -----");  
        displayInfo();  
  
        if (marks >= 85) {  
            System.out.println("Eligible for Merit Scholarship");  
        } else {  
            System.out.println("Not Eligible");  
        }  
    }  
}
```

```
class NeedBasedScholarship extends Scholarship {
```

```
    double familyIncome;
```

```
    NeedBasedScholarship(String name, double marks, double familyIncome) {  
        super(name, marks);    this.familyIncome = familyIncome;  
    }
```

```
    @Override  
    void checkEligibility() {  
        System.out.println("----- Need Based Scholarship -----");  
        displayInfo();  
        System.out.println("Family Income: " + familyIncome);  
  
        if (marks >= 70 && familyIncome < 50000) {  
            System.out.println("Eligible for Need-Based Scholarship");  
        } else {  
            System.out.println("Not Eligible");  
        }  
    }  
}
```

```
public class ScholarshipSystemQ10 {
```

```
    public static void main(String[] args) {
```

```

MeritScholarship m1 = new MeritScholarship("Ram Sharma", 90);
NeedBasedScholarship n1 = new NeedBasedScholarship("Sita Rai", 75, 30000);

m1.checkEligibility();
System.out.println();
n1.checkEligibility();
}}

```

Output

```

Run ScholarshipSystemQ10 x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Fi
----- Merit Scholarship -----
Name: Ram Sharma
Marks: 90.0
Eligible for Merit Scholarship

----- Need Based Scholarship -----
Name: Sita Rai
Marks: 75.0
Family Income: 30000.0
Eligible for Need-Based Scholarship

Process finished with exit code 0

```

8) FreeSystemQ4.java

```

abstract class StudentFee {
String name;
    int id;

    StudentFee(String name, int id) {
        this.name = name;
this.id = id;
    }

    abstract double calculateFee();

    void displayInfo() {
        System.out.println("Name: " + name);
        System.out.println("ID: " + id);
    }
}

class Undergraduate extends StudentFee {

```

```

    Undergraduate(String name, int id) {
        super(name, id);
    }

    @Override double
    calculateFee() {
        return 50000; // example fee
    }

    @Override void
    displayInfo() {
        System.out.println("----- Undergraduate Student -----");
        super.displayInfo();
        System.out.println("Fee: " + calculateFee());
    }
}

class Graduate extends StudentFee {

    Graduate(String name, int id) {
        super(name, id);
    }

    @Override double
    calculateFee() {
        return 80000; // example fee
    }

    @Override void
    displayInfo() {
        System.out.println("----- Graduate Student -----");
        super.displayInfo();
        System.out.println("Fee: " + calculateFee());
    }
}

public class FeeSystemQ4 {
    public static void main(String[] args) {

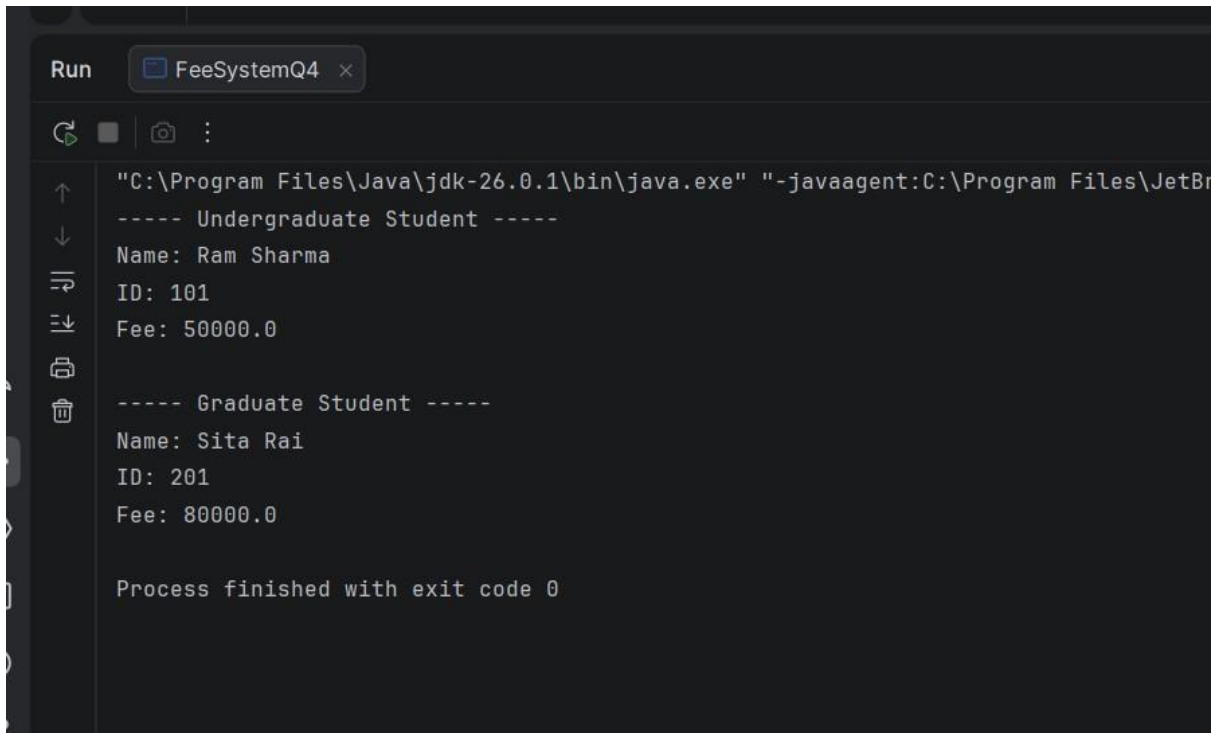
        Undergraduate u1 = new Undergraduate("Ram Sharma", 101);
        Graduate g1 = new Graduate("Sita Rai", 201);

        u1.displayInfo();
        System.out.println();

        g1.displayInfo();
    }
}

```

Output



```
Run  FeeSystemQ4 x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBr
----- Undergraduate Student -----
Name: Ram Sharma
ID: 101
Fee: 50000.0

----- Graduate Student -----
Name: Sita Rai
ID: 201
Fee: 80000.0

Process finished with exit code 0
```

9) ResultSystemQ5.java

```
interface ResultProcessor {
    void calculateGrade(int marks);
}

class EngineeringDepartment implements ResultProcessor {

    @Override
    public void calculateGrade(int marks) {
        System.out.println("----- Engineering Department -----");

        if (marks >= 80) {
            System.out.println("Grade: A");
        } else if (marks >= 60) {
            System.out.println("Grade: B");
        } else if (marks >= 40) {
            System.out.println("Grade: C");
        } else {
            System.out.println("Grade: F");
        }
    }
}
```

```
}  
}
```

```
class ManagementDepartment implements ResultProcessor {
```

```
    @Override
```

```
    public void calculateGrade(int marks) {
```

```
        System.out.println("----- Management Department -----");
```

```
        if (marks >= 85) {
```

```
            System.out.println("Grade: A");
```

```
        } else if (marks >= 70) {
```

```
            System.out.println("Grade: B");
```

```
        } else if (marks >= 50) {
```

```
            System.out.println("Grade: C");
```

```
        } else {
```

```
            System.out.println("Grade: F");
```

```
        }
```

```
    }
```

```
}
```

```
public class ResultSystemQ5 {
```

```
    public static void main(String[] args) {
```

```
        EngineeringDepartment eng = new EngineeringDepartment();
```

```
        ManagementDepartment mgmt = new ManagementDepartment();
```

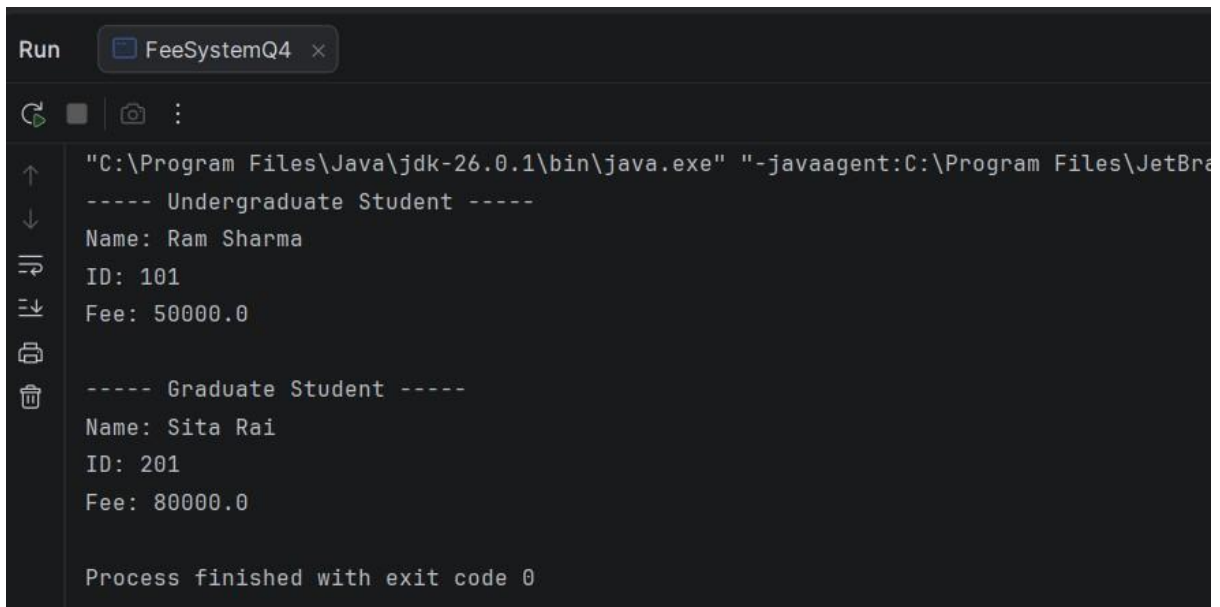
```
        eng.calculateGrade(78);
```

```
        System.out.println();
```

```
        mgmt.calculateGrade(78);
```

```
    } }
```

Output



```
Run  FeeSystemQ4 x
"\"C:\Program Files\Java\jdk-26.0.1\bin\java.exe\" \"-javaagent:C:\Program Files\JetBra
----- Undergraduate Student -----
Name: Ram Sharma
ID: 101
Fee: 50000.0

----- Graduate Student -----
Name: Sita Rai
ID: 201
Fee: 80000.0

Process finished with exit code 0
```

10) LibrarySystemQ6.java

```
class Member {
    String name;

    Member(String name) {
        this.name = name;
    }

    int getBorrowLimit() {
        return 0;
    }

    void displayInfo() {
        System.out.println("Name: " + name);
    }
}

class StudentMember extends Member {

    StudentMember(String name) {
        super(name);
    }
}
```



```

        @Override    int
getBorrowLimit() {
    return 3;
}

        @Override    void
displayInfo() {
    System.out.println("----- Student Member -----");
    super.displayInfo();
    System.out.println("Borrow Limit: " + getBorrowLimit());
}
}

```

```

class TeacherMember extends Member {

```

```

    TeacherMember(String name) {
        super(name);
    }

        @Override    int
getBorrowLimit() {
    return 10;
}

        @Override    void
displayInfo() {
    System.out.println("----- Teacher Member -----");
    super.displayInfo();
    System.out.println("Borrow Limit: " + getBorrowLimit());
}
}

```

```

public class LibrarySystemQ6 {
    public static void main(String[] args) {

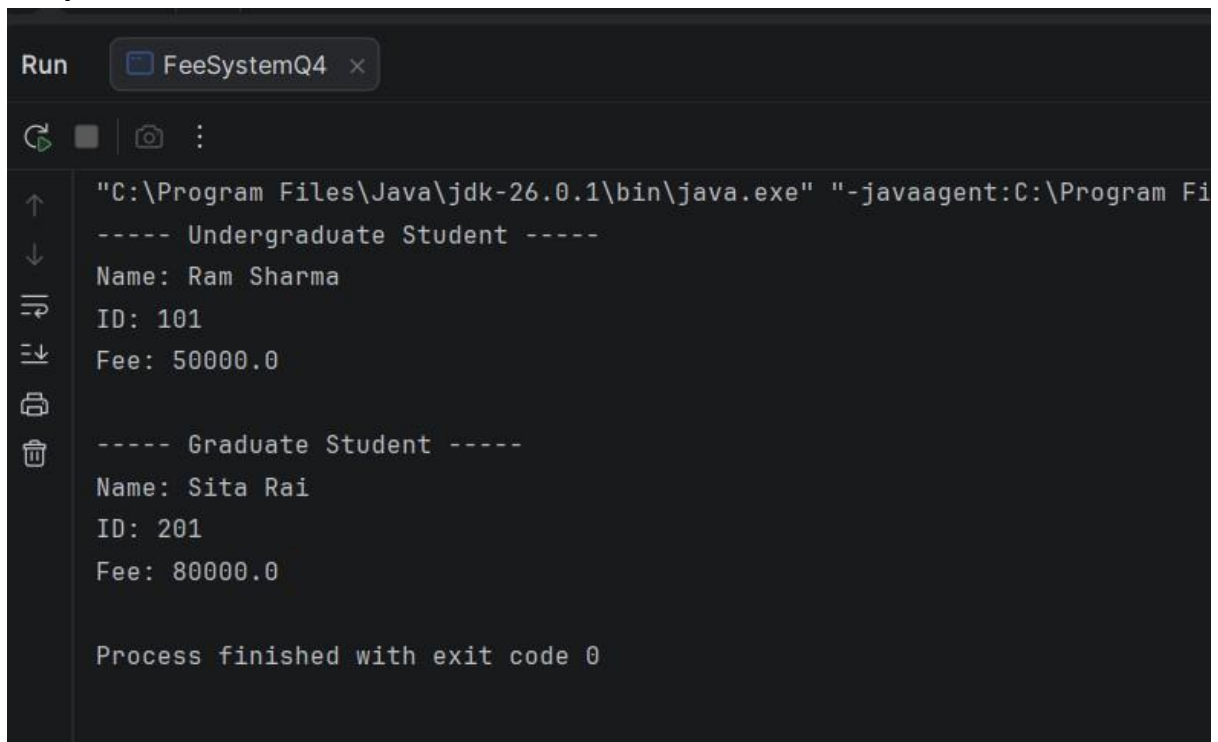
        StudentMember s1 = new StudentMember("utsav");
        TeacherMember t1 = new TeacherMember("");

        s1.displayInfo();
        System.out.println();

        t1.displayInfo();
    }
}

```

Output



```
Run FeeSystemQ4 x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Fi
----- Undergraduate Student -----
Name: Ram Sharma
ID: 101
Fee: 50000.0

----- Graduate Student -----
Name: Sita Rai
ID: 201
Fee: 80000.0

Process finished with exit code 0
```